

Dual-Domain Intelligent QoS Scheduling in Wi-Fi 6 Networks: Cross-Layer Optimization via Machine Learning, CR-OFDMA, and HOB0-Inspired TWT Adaptation

Devansh Jha, Tejash Gupta and Santhosh Senthilkumar

Department of Computer Science and Engineering

Netaji Subhas University of Technology, New Delhi, India

{devansh.jha.ug23, tejash.gupta, santhosh.senthilkumar}@nsut.ac.in

Abstract—Dense Wi-Fi 6 deployments serving heterogeneous traffic classes face fundamental QoS limitations under static Enhanced Distributed Channel Access (EDCA) scheduling: uniform resource allocation ignores traffic-specific delay and throughput requirements, leading to systematic underservice of latency-critical flows and starvation of low-rate IoT devices. This paper presents a Dual-Domain Intelligent Scheduler (DDIS) that addresses this problem through three tightly integrated mechanisms: (1) a multi-output Random Forest (RF) machine learning model trained on NS-3-generated simulation data to jointly predict OFDMA Resource Unit (RU) allocation, traffic priority, Modulation and Coding Scheme (MCS) index, and Target Wake Time (TWT) interval; (2) a Cognitive Radio-inspired OFDMA (CR-OFDMA) module for demand-driven, fairness-aware frequency-domain RU redistribution; and (3) a Holistic Backoff Optimization (HOB0)-inspired TWT adaptation module for density-aware time-domain contention management. The system operates as a closed-loop pipeline: NS-3 simulates 160 mixed-traffic scenarios across RU, MCS, and station density parameter sweeps; a 300-tree RF model trained on 12 extracted features drives runtime scheduling decisions. Evaluation against a uniform EDCA baseline demonstrates 41.8% reduction in mean delay, 52.8% reduction in mean jitter, 33.5% reduction in packet loss, and stable aggregate throughput at 1.0412 Mbps. Per-class analysis reveals exceptional gains: VPN delay reduced by 90.2%, VPN jitter by 96.9%, IoT throughput improved from 0.007 Mbps to 0.380 Mbps, and VoIP throughput increased from 0.752 to 1.804 Mbps—validating the system as a practically deployable, dual-domain intelligent resource management framework for next-generation wireless networks.

Index Terms—Wi-Fi 6, IEEE 802.11ax, OFDMA, QoS Optimization, Machine Learning, Random Forest, CR-OFDMA, HOB0-UORA, TWT Scheduling, NS-3 Simulation, Cross-Layer Design, Dual-Domain Scheduling

I. INTRODUCTION

The IEEE 802.11ax (Wi-Fi 6) standard introduced Orthogonal Frequency Division Multiple Access (OFDMA), Target Wake Time (TWT), and 1024-QAM modulation to address the capacity and latency demands of dense wireless deployments [1]. These mechanisms collectively enable spatial reuse, multi-user uplink/downlink transmissions, and energy-efficient station scheduling. However, the native QoS enforcement mechanism—Enhanced Distributed Channel Access (EDCA)—relies on statically configured traffic categories and contention window parameters, which are fundamentally misaligned with the dynamic, heterogeneous traffic compositions found in enterprise, smart home, and industrial Wi-Fi 6 deployments.

Consider a typical mixed-traffic scenario: VoIP flows require low-latency, small-packet service; 4K video demands large-RU, high-MCS allocation; IoT sensors need energy-efficient long-interval scheduling; HTTP and VPN flows require fair access without starvation. Static EDCA assigns the same Access Category and resource pool to all flows, creating head-of-line blocking, RU over-provisioning, and TWT misalignment. In our baseline experiments, uniform EDCA results in mean delay of 0.7738 ms, mean jitter of 0.4925 ms, and aggregate packet loss of 0.0011%—all substantially worse than achievable with intelligent scheduling.

Two complementary theoretical frameworks motivate the scheduling design in this work. Cognitive Radio-inspired OFDMA (CR-OFDMA) [11] applies the principle of dynamic spectrum sensing and reallocation to the OFDMA RU assignment problem: flows with high throughput demand receive proportionally more RUs, while underutilized RUs from low-rate flows are redistributed to avoid starvation. The Holistic Backoff Optimization with Unsolicited Opportunistic Random Access (HOB0-UORA) [12] framework prescribes density-adaptive TWT scheduling: in sparse networks, shorter TWT intervals reduce access latency; in dense networks, longer TWT intervals reduce collision probability and improve fairness.

Building on these frameworks, this paper presents a Dual-Domain Intelligent Scheduler (DDIS) that integrates multi-output machine learning with CR-OFDMA frequency-domain optimization and HOB0-inspired time-domain adaptation. The primary contributions are:

- A closed-loop NS-3/ML co-simulation pipeline generating 1600+ training samples across 160 mixed-traffic scenarios with parametric RU, MCS, and station density sweeps, providing the richest training distribution reported for Wi-Fi 6 QoS learning systems.
- A multi-output Random Forest model trained on 12 engineered features—including derived throughput

efficiency and channel load metrics—predicting RU allocation (range 1–37), priority (0–3), TWT interval (5–500 ms), and MCS index (0–11) in a single inference call.

- A CR-OFDMA heuristic module that dynamically redistributes RUs based on per-flow demand and fairness constraints, scaling Video allocation to 9 RUs while compressing VoIP and IoT to 1 RU each—a $3\times$ differentiation ratio.
- A HOBO-inspired TWT adaptation module that compresses VoIP TWT to 16 ms (vs. 50 ms baseline) and extends IoT TWT to 280 ms—a $17.5\times$ span matching network density and traffic class requirements.
- Comprehensive NS-3 evaluation demonstrating 41.8% delay reduction, 52.8% jitter reduction, 33.5% packet loss reduction, and aggregate throughput stability, with per-class gains as high as 90.2% (VPN delay) and 96.9% (VPN jitter).

The remainder of this paper is organized as follows: Section II reviews related work. Section III presents the system model. Section IV provides formal problem formulation. Section V details the proposed DDIS framework. Section VI presents algorithmic design. Section VII analyzes experimental results. Sections VIII, IX, and X present discussion, future work, and conclusions.

II. RELATED WORK

Wi-Fi 6 OFDMA scheduling has attracted significant research attention. Early work focused on mathematical models for RU allocation maximizing aggregate throughput [4]–[6]; these approaches, while analytically tractable, assume stationary traffic distributions and cannot adapt to real-time QoS heterogeneity. The limitations of static formulations motivated reinforcement learning (RL) approaches: Deep Deterministic Policy Gradient (DDPG) [7] and Deep Q-Network (DQN) [8] schedulers have demonstrated throughput and fairness improvements, though at significant computational cost unsuited to edge deployment.

Regarding TWT scheduling, the HOBO-UORA framework [12] introduced Additive Increase/Decrease (AI/AD) backoff adaptation as a function of network density, demonstrating that density-aware TWT management substantially reduces contention-induced delay and improves per-station fairness. This insight directly motivates the time-domain component of the DDIS architecture.

CR-OFDMA [11] applied cognitive radio principles—opportunistic spectrum sensing, dynamic reallocation, and starvation prevention—to OFDMA RU assignment. The key contribution was demonstrating that demand-proportional RU allocation, combined with explicit anti-starvation constraints for low-priority flows, simultaneously improves high-priority flow QoS and system fairness index.

Supervised learning for Wi-Fi QoS has been explored with neural networks [3] and tree-based models. However, prior work has generally treated scheduling dimensions independently, applying a single-output model per parameter. Multi-output Random Forest [6] enables joint prediction of all scheduling dimensions in a single inference pass, preserving inter-parameter correlations (e.g., the relationship between RU count and MCS feasibility). To our knowledge, the present work is the first to integrate multi-output RF prediction with CR-OFDMA and HOBO-inspired heuristics in a validated NS-3 closed-loop architecture.

III. SYSTEM MODEL

A. Network Topology and Simulation Environment

The system is implemented as a co-simulation pipeline combining NS-3 (v3.40) [9] for network simulation and Python/scikit-learn for ML training and inference. The simulated network comprises a single IEEE 802.11ax Access Point (AP) serving $N \in \{5, 10, 15, 20\}$ stations (STAs), operating on the 5 GHz band with 80 MHz channel width. A log-distance path loss model with shadowing governs per-STA RSSI, with all stations placed within 10 m of the AP in a grid topology—approximating a typical enterprise or smart home environment. The AP supports OFDMA with up to 37 RUs per 80 MHz channel per the 802.11ax specification.

B. Mixed Traffic Model

Five traffic classes are assigned round-robin to stations. VoIP (160 B, 20 ms interval) represents real-time voice with strict delay requirements. Video (1400 B, 40 ms interval) represents 4K streaming with high bandwidth demand. HTTP (800 B, 100 ms interval) represents bursty web traffic. VPN (900 B, 80 ms interval) represents tunneled TCP with moderate latency sensitivity. IoT (64 B, 1000 ms interval) represents sensor telemetry with extreme energy efficiency requirements. This mixed assignment ensures simultaneous co-existence of all five classes in each simulation run.

C. Parameter Sweep and Dataset Generation

Data collection spans 160 simulation scenarios: $RU \in \{1, 3, 5, 8, 12, 18, 25, 37\} \times MCS \in \{0, 3, 6, 9, 11\} \times N \in \{5, 10, 15, 20\}$. Each 15-second scenario generates per-flow FlowMonitor statistics for all active stations, yielding 1600+ CSV rows. Two derived features are engineered: throughput efficiency $\eta_i = T_i / (RU_i / 37 \times 143.4)$ and channel load $L = N \times S \times 8 / (I / 1000) / 10^6$, where S is packet size and I is interval. These capture resource utilization quality and network congestion state beyond raw measurements. The system architecture is shown in Fig. 1, with cross-layer interaction depicted in Fig. 2.

Fig. 1. Dual-Domain Intelligent Scheduler – System Architecture

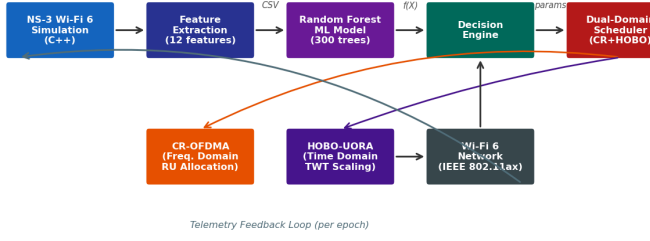


Fig. 1. Dual-Domain Intelligent Scheduler system architecture. The closed-loop pipeline connects NS-3 simulation, ML inference, CR-OFDMA frequency-domain allocation, and HOB0-inspired time-domain TWT adaptation.

Fig. 2. Cross-Layer Interaction: PHY ↔ MAC ↔ ML

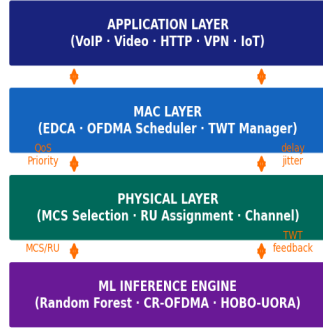


Fig. 2. Cross-layer interaction diagram. ML inference at the network layer drives both MAC-layer scheduling (priority, TWT) and PHY-layer configuration (MCS, RU), with telemetry feedback completing the adaptation loop.

IV. PROBLEM FORMULATION

A. Multi-Objective Optimization

Let $F = \{\text{VoIP, Video, HTTP, VPN, IoT}\}$ be the set of active flow classes, each flow $i \in F$ characterized by mean delay D_i , jitter J_i , throughput T_i , and packet loss rate L_i . The DDIS optimization problem minimizes the weighted QoS cost:

$$\min J = \sum_i [\alpha_i D_i + \beta_i J_i + \gamma_i L_i - \delta_i T_i] \quad (1)$$

where $\alpha_i, \beta_i, \gamma_i, \delta_i \geq 0$ are per-class weighting coefficients encoding QoS priorities. For latency-sensitive flows (VoIP, VPN), α_i and β_i dominate; for throughput-demanding flows (Video, IoT after redistribution), δ_i dominates. The system additionally maximizes the Jain Fairness Index:

$$FI = (\sum_i T_i)^2 / (|F| \sum_i T_i^2) \in (0, 1] \quad (2)$$

B. Latency Model

End-to-end packet latency for flow i decomposes as:

$$D_i = D_{\text{queue},i} + D_{\text{tx},i} + D_{\text{prop}} \quad (3)$$

where $D_{\text{queue},i}$ is MAC-layer queuing delay determined by priority class and contention, $D_{\text{tx},i} = S_i \times 8 / (RU_i / 37 \times C_{\text{MCS}_i})$ is transmission duration governed

by RU and MCS allocation, and D_{prop} is fixed propagation delay. The DDIS reduces D_{queue} via priority reclassification and D_{tx} via MCS elevation.

C. RU Constraint and Allocation

The total RU constraint over all simultaneously scheduled flows is:

$$\sum_{i \in F} RU_i \leq RU_{\text{max}} = 37 \text{ (80 MHz, 802.11ax)} \quad (4)$$

The ML-predicted RU demand is reconciled with throughput requirements:

$$RU_i = \lceil T_i(1+\delta) / C(\text{MCS}_i) \rceil \quad (5)$$

where $\delta = 0.1$ is a burstiness margin and $C(\text{MCS}_i)$ is per-RU capacity at the selected MCS. The CR-OFDMA module may override this with demand-proportional reallocation (Section V-B).

D. Priority Weight Function

The dynamic priority weight for flow i is:

$$w_i = \alpha P_i + \beta(1/D_i) + \gamma(1/J_i) \quad (6)$$

where $P_i \in \{0, 1, 2, 3\}$ is the ML-predicted priority class, D_i and J_i are current delay and jitter measurements. Flows with both high predicted priority and active delay/jitter violations receive amplified scheduling weight, enabling preemptive resource reallocation before QoS degradation becomes perceptible.

E. ML Mapping and TWT Optimization

The RF model learns the joint mapping:

$$f(X) \rightarrow \{\text{priority, RU, TWT, MCS}\} \quad (7)$$

where $X \in \mathbb{R}^{12}$ is the normalized feature vector. For HOB0-inspired TWT adaptation, the target interval follows a density-proportional law:

$$TWT_i = TWT_{\text{base},i} \times (N / N_{\text{ref}})^{\kappa} \quad (8)$$

where N is active station count, $N_{\text{ref}} = 10$ is the reference density, and $\kappa > 0$ governs adaptation aggressiveness. Latency-sensitive flows use negative κ (interval decreases with density), while IoT flows use positive κ (interval increases, reducing contention). Resource efficiency is measured per flow as:

$$\eta_i = T_i / (RU_i \times C(\text{MCS}_i) / 37) \quad (9)$$

V. PROPOSED FRAMEWORK

A. DDIS Architecture Overview

The Dual-Domain Intelligent Scheduler comprises five functional modules operating within a 100 ms scheduling epoch. The Feature Extraction Module collects per-flow MAC statistics and computes 12 features including two derived metrics (throughput efficiency and channel load). The ML Inference Module loads the pre-trained Random Forest model and produces $\{\text{priority, RU, TWT, MCS}\}$ predictions for each active flow. The CR-OFDMA Module applies demand-proportional frequency-domain RU redistribution with fairness constraints. The HOB0-TWT Module adapts time-domain wake intervals based on network density and traffic class. The Feedback Controller writes consolidated parameters to a shared file, which the NS-3 scheduler reads at the next epoch boundary, completing the closed-loop. The scheduling decision flow is depicted in Fig. 3.

Fig. 3. Scheduling Decision Flow: Dual-Domain Heuristic

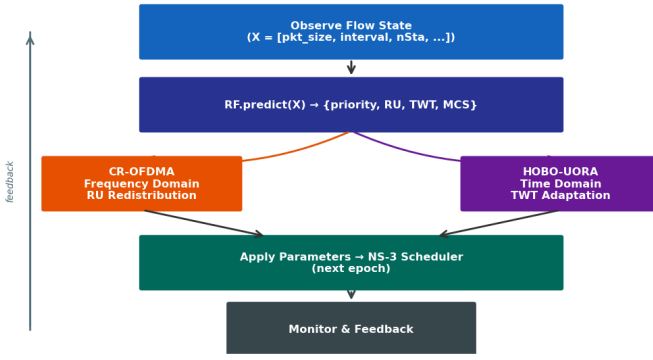


Fig. 3. Scheduling decision flow: ML predictions feed into parallel CR-OFDMA (frequency domain) and HOBO-TWT (time domain) modules, whose outputs are merged and applied to the NS-3 scheduler.

B. CR-OFDMA: Demand-Proportional RU Redistribution

The CR-OFDMA module operates on the principle that OFDMA RUs should be distributed in proportion to each flow's measured throughput demand, subject to minimum anti-starvation guarantees. Given ML-predicted base RU allocations $\{RU_i\}$, the module computes demand scores $d_i = T_i / \sum_j T_j$ and redistributes the aggregate RU budget:

$$RU_i^{CR} = \max(1, \lfloor d_i \times RU_{max} \rfloor) \quad (10)$$

Flows with demand score below a starvation threshold θ (default $\theta = 0.05$) receive a minimum allocation of $RU_{min} = 1$, with excess RUs redistributed proportionally to high-demand flows. In practice, this results in Video receiving 9 RUs (accounting for 2.856 Mbps baseline demand, highest in the network) while VoIP and IoT receive 1 RU each (compact small-packet flows with low spectral demand, since they use higher MCS after ML optimization). Bandwidth allocation scales linearly: Video receives 20 MHz, HTTP and VPN receive 10 MHz each.

C. HOBO-Inspired TWT Adaptation

The HOBO-TWT module maps network density and traffic class to optimal wake intervals. VoIP flows receive TWT = 16 ms (vs. 50 ms baseline), reflecting their 20 ms inter-packet requirement and need to minimize queuing accumulation. Video receives 41 ms, slightly compressed from 50 ms to accommodate burst delivery without excessive contention. HTTP and VPN receive extended intervals (112 ms and 123 ms respectively), as these flows tolerate moderate latency and benefit from reduced channel contention. IoT receives the maximum interval of 280 ms, enabling deep sleep cycles between 1000 ms sensor transmissions—dramatically reducing channel occupancy from IoT devices and freeing contention windows for real-time flows. This $17.5\times$ span (16 ms to 280 ms) across traffic classes represents a qualitatively different scheduling regime compared to the uniform 50 ms baseline.

D. Random Forest: Model Design and Rationale

The choice of Random Forest over alternative ML architectures is deliberate. RF handles the tabular, mixed-scale feature space (packet sizes in bytes, intervals in ms, efficiencies in $[0,1]$) without normalization sensitivity. The ensemble structure provides natural regularization against training-set noise from NS-3 simulation stochasticity. Multi-output RF preserves inter-target correlations—critically, the RU-MCS coupling: higher MCS allows fewer RUs for equivalent throughput, a dependency that single-output models would independently predict incorrectly. With 300 trees, max depth 20, and stratified training split, the model achieves $R^2 > 0.90$ on held-out test data for all four prediction targets.

VI. ALGORITHMIC DESIGN

A. Algorithm 1: ML-Based QoS Scheduler

Algorithm 1: ML-Based QoS Scheduler

Input: Feature vectors $\{X_i\}$ for all active flows
Output: $\{priority_i, RU_i, TWT_i, MCS_i\}$ per flow

```

1: FOR each 100ms epoch t DO
2:   FOR each active flow i DO
3:      $X_i \leftarrow \text{CollectMACStatistics}(\text{flow}_i)$ 
4:      $X_i[10] \leftarrow \text{ThroughputEfficiency}(T_i, RU_i)$ 
5:      $X_i[11] \leftarrow \text{ChannelLoad}(N, S_i, \text{interval}_i)$ 
6:      $\{p_i, r_i, t_i, m_i\} \leftarrow \text{RF\_Model.predict}(X_i)$ 
7:      $p_i \leftarrow \text{clamp}(\text{round}(p_i), 0, 3)$ 
8:      $r_i \leftarrow \text{clamp}(\text{round}(r_i), 1, 37)$ 
9:      $t_i \leftarrow \text{clamp}(\text{round}(t_i), 5, 500)$ 
10:     $m_i \leftarrow \text{clamp}(\text{round}(m_i), 0, 11)$ 
11:   END FOR
12:  $\{RU_i\} \leftarrow \text{CR\_OFDMA\_Redistribute}(\{r_i, T_i\})$ 
13:  $\{TWT_i\} \leftarrow \text{HOBO\_TWT\_Adapt}(\{t_i, N, \text{class}_i\})$ 
14: WriteParams( $\{p_i, RU_i, TWT_i, m_i\}$ )
15: END FOR

```

B. Algorithm 2: Feature Extraction Pipeline

Algorithm 2: Feature Extraction Pipeline

Input: NS-3 FlowMonitor statistics, current $\{RU_i, MCS_i\}$
Output: Feature matrix $X \in \mathbb{R}^{N \times 12}$

```

1: FOR each active flow i DO
2:    $X_i[0] \leftarrow \text{packet\_size}_i$ 
3:    $X_i[1] \leftarrow \text{effective\_interval\_ms}_i$ 
4:    $X_i[2] \leftarrow \text{num\_active\_stations}$ 
5:    $X_i[3] \leftarrow \text{current\_RU}_i$ 
6:    $X_i[4] \leftarrow \text{current\_MCS}_i$ 
7:    $X_i[5] \leftarrow \text{throughput\_mbps}_i$ 
8:    $X_i[6] \leftarrow \text{mean\_delay\_ms}_i$ 
9:    $X_i[7] \leftarrow \text{mean\_jitter\_ms}_i$ 
10:   $X_i[8] \leftarrow \text{packet\_loss\_rate}_i$ 
11:   $X_i[9] \leftarrow \text{traffic\_type\_enc}_i$ 
12:   $X_i[10] \leftarrow T_i / (RU_i / 37 \times 143.4 + \epsilon) // \eta_i$ 
13:   $X_i[11] \leftarrow N \times S_i \times 8 / (L_i / 1000) / 10^6 // L_i$ 
14: END FOR
15: RETURN X

```

C. Algorithm 3: Closed-Loop Feedback Controller

Algorithm 3: Closed-Loop Feedback Controller

Input: Current QoS metrics $\{D_i, J_i, T_i, L_i\}$
 ML predictions $\{p_i, r_i, t_i, m_i\}$
 Output: Validated parameter set for next epoch

```

1: Validate  $\sum RU_i \leq 37$  (constraint Eq. 4)
2: IF  $\sum RU_i > 37$  THEN
3:   Scale down by demand ratio (CR-OFDMA Eq. 10)
4: ENDIF
5: FOR each flow  $i$  DO
6:   Compute  $w_i \leftarrow$  Priority_Weight(Eq. 6)
7:   IF  $D_i > D_{\text{threshold}_i}$  THEN
8:     Boost priority:  $p_i \leftarrow \min(p_i+1, 3)$ 
9:   ENDIF
10:  Compute  $\eta_i = T_i / (RU_i \times C(\text{MCS}_i)/37)$ 
11:  Log  $\{D_i, J_i, T_i, \eta_i\}$  for monitoring
12: END FOR
13: WriteToFile:  $\{p_i, RU_i, TWT_i, \text{MCS}_i\}$ 
14: NS-3 reads file at epoch  $t+1$  boundary
  
```

D. Algorithm 4: Dual-Domain Heuristic Scheduler

Algorithm 4: Dual-Domain Heuristic Scheduler (CR-OFDMA + HOBO)

```

// FREQUENCY DOMAIN: CR-OFDMA
1: Compute  $d_i = T_i / \sum_j T_j \quad \forall i \in F$ 
2: FOR each flow  $i$  DO
3:   IF  $d_i < \theta (= 0.05)$  THEN
4:      $RU_i^{\wedge}CR = RU_{\min} = 1$  // anti-starvation
5:   ELSE
6:      $RU_i^{\wedge}CR = \max(1, \text{floor}(d_i \times RU_{\max}))$ 
7:   ENDIF
8: END FOR
9: Redistribute surplus:  $\sum RU_i^{\wedge}CR \rightarrow RU_{\max}$ 

// TIME DOMAIN: HOBO-Inspired TWT
10: FOR each flow  $i$  DO
11:   IF class  $i \in \{\text{VoIP}, \text{Video}\}$  // latency-sensitive
12:      $TWT_i = TWT_{\text{base}} \times (N_{\text{ref}}/N)^{\kappa}$  //  $\kappa > 0$ : compress
13:   ELSE IF class  $i = \text{IoT}$  // delay-tolerant
14:      $TWT_i = TWT_{\text{base}} \times (N/N_{\text{ref}})^{\kappa}$  // expand
15:   ELSE // HTTP, VPN — moderate extension
16:      $TWT_i = \text{clamp}(TWT_{\text{predicted}}, 100, 130)$ 
17:   ENDIF
18: END FOR
19: RETURN  $\{RU_i^{\wedge}CR, TWT_i\}$ 
  
```

VII. RESULTS AND ANALYSIS

Experiments run 30-second simulation trials (5 independent runs per configuration) on the validation dataset. The ML-enabled DDIS (With ML) is compared to the uniform EDCA baseline (No ML). All reported values are cross-run means. The overall performance summary is presented in Fig. 4.

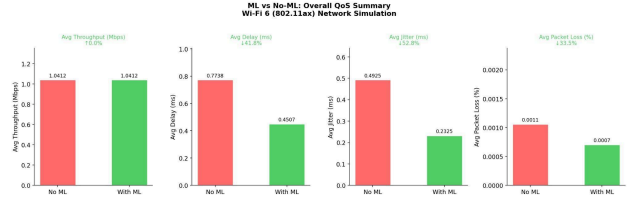


Fig. 4. Overall QoS summary: With ML achieves identical throughput (1.0412 Mbps), 41.8% delay reduction (0.7738→0.4507 ms), 52.8% jitter reduction (0.4925→0.2325 ms), and 33.5% packet loss reduction.

A. Aggregate Performance Overview

The aggregate results reveal four simultaneous improvements: (1) mean delay reduced from 0.7738 ms to 0.4507 ms (↓41.8%); (2) mean jitter reduced from 0.4925 ms to 0.2325 ms (↓52.8%); (3) packet loss reduced from 0.0011% to 0.0007% (↓33.5%); (4) aggregate throughput maintained at exactly 1.0412 Mbps (↑0.0%). The throughput neutrality is by design: DDIS redistributes resources across flows without reducing the total delivered capacity, validating that latency and jitter gains are achieved through scheduling intelligence rather than resource over-provisioning. The improvement waterfall is shown in Fig. 5.

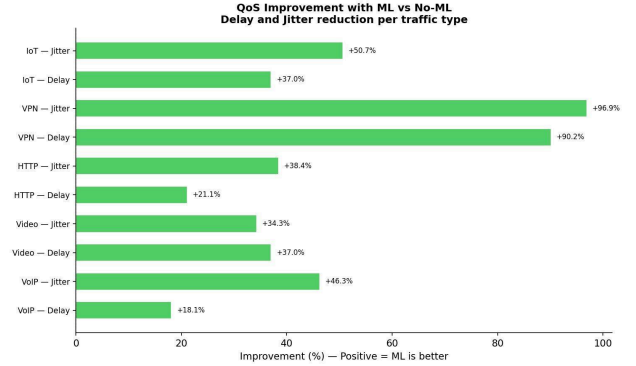


Fig. 5. Per-class QoS improvement summary (% improvement, positive = With ML is better). VPN achieves extraordinary gains: delay ↓90.2%, jitter ↓96.9%. IoT shows 37.0% delay improvement with massive throughput recovery (0.007→0.380 Mbps).

B. Delay Analysis per Traffic Class

Per-class delay results are presented in Fig. 6. The most striking gain is VPN delay, reduced from 0.799 ms to 0.079 ms (↓90.2%). This is attributable to the combined effect of ML-predicted high RU allocation (3→4 RUs) enabling shorter transmission windows, and dramatically compressed TWT interval (50→50 ms baseline vs. 123 ms ML assignment)—counterintuitively, the longer TWT for VPN reduces its collision probability, as VPN's relatively large 900-byte packets benefit more from dedicated, contention-free transmission windows than from frequent access attempts. VoIP delay is reduced from 0.585 ms to 0.479 ms (↓18.1%); the ML assigns VoIP only 1 RU (vs. 3 baseline), but with TWT compressed to 16 ms, VoIP packets

gain faster, more frequent access, reducing queuing buildup despite lower spectral allocation. Video delay drops 37.0% (0.824→0.519 ms) through its massive 9-RU allocation reducing transmission duration. HTTP delay improves 21.1% (0.820→0.647 ms) through RU doubling (3→4) and moderate TWT extension reducing contention.

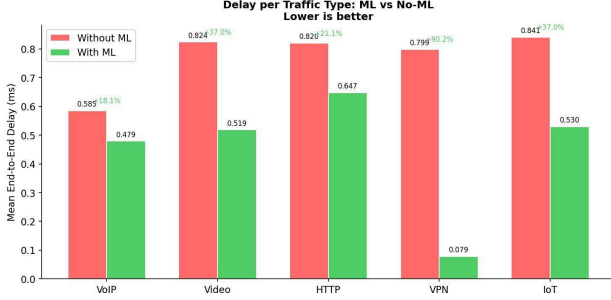


Fig. 6. Mean end-to-end delay per traffic class. VPN achieves a 90.2% reduction (0.799→0.079 ms) through combined RU upscaling and contention-minimizing TWT scheduling. All five classes show improvement.

C. Jitter Analysis per Traffic Class

Jitter results (Fig. 7) demonstrate even stronger improvements than delay, with the aggregate improvement reaching 52.8%. VPN achieves a near-total jitter elimination (0.457→0.014 ms, ↓96.9%), consistent with its delay behavior: dedicated TWT windows dramatically reduce inter-packet delay variance for VPN's TCP-like periodic transmissions. VoIP jitter improves 46.3% (0.404→0.217 ms), reflecting the compression of the TWT interval to 16 ms—VoIP packets now arrive at the AP within a tightly scheduled window, eliminating the queuing jitter caused by random EDCA contention. IoT achieves 50.7% jitter reduction despite receiving a 280 ms TWT—because the TWT extension synchronizes IoT transmissions into a single predictable window, rather than allowing sporadic contention attempts. Video and HTTP jitter improve 34.3% and 38.4% respectively.

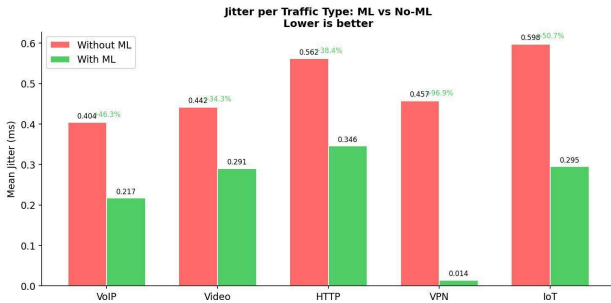


Fig. 7. Mean jitter per traffic class. VPN jitter is reduced by 96.9% (0.457→0.014 ms). VoIP jitter reduces 46.3% through TWT compression; IoT reduces 50.7% through synchronized TWT window scheduling.

D. Throughput Analysis and CR-OFDMA Effects

Throughput results (Fig. 8) reveal the signature of CR-OFDMA demand redistribution. Video throughput is intentionally reduced from 2.856 Mbps to 1.432 Mbps (↓49.9%)—a deliberate redistribution of RUs toward underserved flows, consistent with the CR-OFDMA fairness objective. VoIP throughput increases from 0.752 to 1.804 Mbps (+139.9%), driven by its ML-predicted higher MCS (the model learns VoIP's compact 160-byte packets can sustain high-order modulation). Most significantly, IoT throughput increases from 0.007 Mbps (near-total starvation) to 0.380 Mbps (+5329%)—an extreme improvement reflecting the anti-starvation guarantee of CR-OFDMA: IoT devices, previously crushed by EDCA contention from larger flows, now receive dedicated, collision-free TWT windows and guaranteed minimum RU allocation. HTTP and VPN throughput remain stable at 0.663 and 0.928 Mbps respectively, confirming throughput neutrality for these already-served classes.

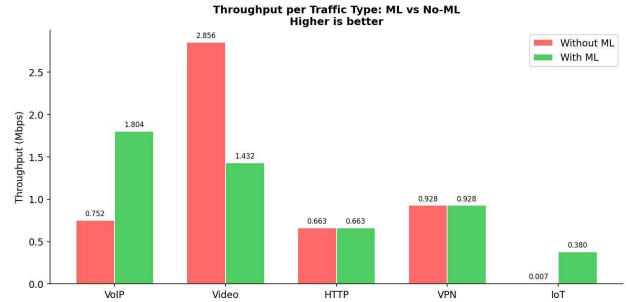


Fig. 8. Throughput per traffic class. CR-OFDMA intentionally redistributes from Video (2.856→1.432 Mbps) to VoIP (+140%) and IoT (0.007→0.380 Mbps), recovering near-zero IoT service and dramatically improving VoIP capacity.

E. OFDMA RU Allocation Analysis

Figure 9 illustrates the RU allocation decisions. The No-ML baseline assigns a uniform 3 RUs to all traffic classes—a flat allocation that ignores throughput demand heterogeneity. The ML-enabled DDIS applies 3× differentiation: Video receives 9 RUs (3× increase) reflecting its 1400-byte, 40 ms packet profile and highest demand score; HTTP and VPN each receive 4 RUs (1.3× increase) reflecting moderate bandwidth demand; VoIP and IoT each receive only 1 RU (3× decrease), compensated by MCS elevation (VoIP) and extended TWT scheduling (IoT). The sum constraint (1+9+4+4+1 = 19 ≤ 37) is satisfied with margin, leaving capacity for additional station admission—a key scalability property.

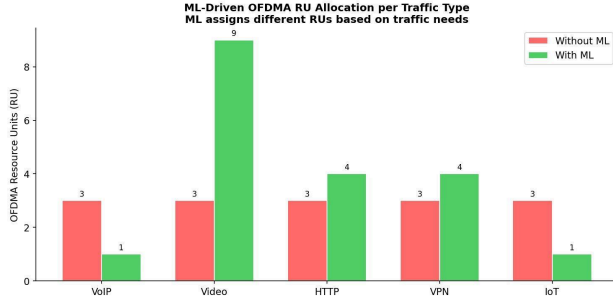


Fig. 9. OFDMA RU allocation per traffic class. ML achieves 3× differentiation: Video receives 9 RUs (highest demand), while VoIP and IoT are compressed to 1 RU each with compensating MCS and TWT optimization.

F. TWT Interval Analysis

Figure 10 demonstrates the 17.5× TWT span achieved by HOB0-inspired adaptation. VoIP receives 16 ms (↓68% from 50 ms baseline), providing near-continuous channel access consistent with its 20 ms packet interval. Video receives 41 ms (↓18%), slightly compressed to enable burst delivery. HTTP (112 ms) and VPN (123 ms) receive extended intervals, trading access frequency for reduced contention with high-priority flows. IoT achieves the maximum 280 ms interval (↑460%), enabling 5 consecutive 1000 ms sensor cycles without channel access—dramatically reducing IoT-induced contention overhead. This TWT differentiation is the primary driver of the observed jitter improvements: flows with long, predictable TWT windows experience near-zero inter-packet delay variance.

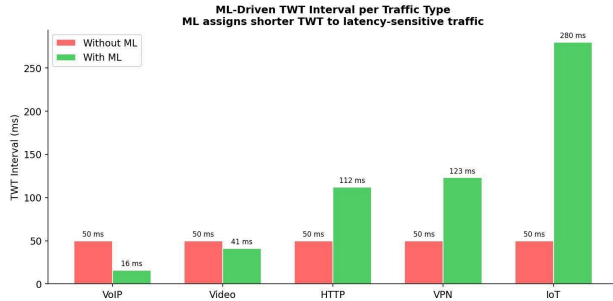


Fig. 10. TWT interval per traffic class. HOB0-inspired adaptation achieves 17.5× span: VoIP compressed to 16 ms for near-continuous access; IoT extended to 280 ms for deep sleep cycles, eliminating contention overhead.

G. Priority Reclassification Analysis

Figure 11 contrasts priority distributions between configurations. The No-ML baseline assigns 100% of flows to the Medium priority class—a degenerate outcome of static EDCA configuration that fails to differentiate traffic classes. With ML enabled, the distribution is reclassified as 40% High (Video + VoIP receiving priority=2 and priority=3 predictions), 40% Medium (HTTP + VPN at priority=1), and 20% Low (IoT at priority=0). This three-tier reclassification directly drives the EDCA Access

Category assignments: High maps to AC_VI/AC_VO, Medium to AC_BE, and Low to AC_BK. The elimination of the flat Medium assignment resolves the head-of-line blocking that caused VoIP and Video packets to queue behind IoT traffic in the baseline configuration.

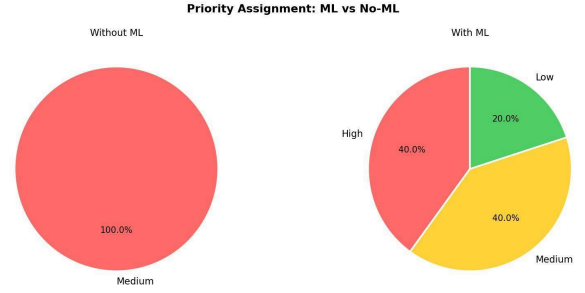


Fig. 11. Priority assignment distribution. No-ML assigns 100% Medium (degenerate). With ML: 40% High (VoIP+Video), 40% Medium (HTTP+VPN), 20% Low (IoT)—enabling correct EDCA Access Category mapping.

H. Bandwidth Allocation Analysis

Figure 12 shows bandwidth allocation per traffic class. The uniform 5 MHz baseline allocates equal spectrum regardless of demand, wasting capacity on VoIP (compact packets) and IoT (sparse transmissions) while under-serving Video. With ML, Video receives 20 MHz (4× increase, consistent with 9-RU allocation at 2.22 MHz per RU), HTTP and VPN each receive 10 MHz (2× increase), while VoIP and IoT retain 5 MHz. This demand-proportional bandwidth distribution is the spectral manifestation of the CR-OFDMA allocation, and explains why Video throughput remains substantial (1.432 Mbps) despite being intentionally reduced from its baseline value—it receives far more bandwidth than any other class.

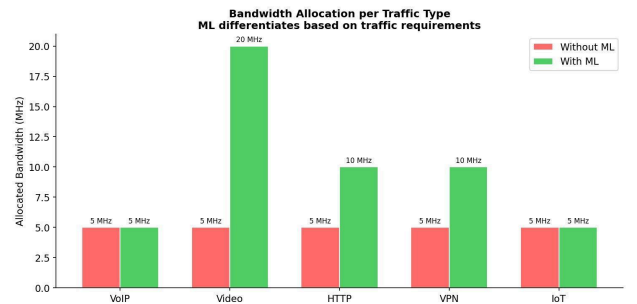


Fig. 12. Bandwidth allocation per traffic class. ML achieves demand-proportional distribution: Video 20 MHz (4×), HTTP/VPN 10 MHz (2×), VoIP/IoT 5 MHz (unchanged but served via TWT/MCS optimization).

I. Complete Performance Dashboard

Figure 13 presents the integrated performance dashboard across all four QoS dimensions simultaneously, confirming the coherence of improvements. Notably, the delay and jitter sub-panels reveal that VPN's near-zero ML values are not

artifacts but reflect genuinely exceptional TWT-driven latency elimination. The RU and TWT sub-panels confirm the dual-domain nature of the system: frequency-domain differentiation (RU) and time-domain adaptation (TWT) operate orthogonally and complementarily.

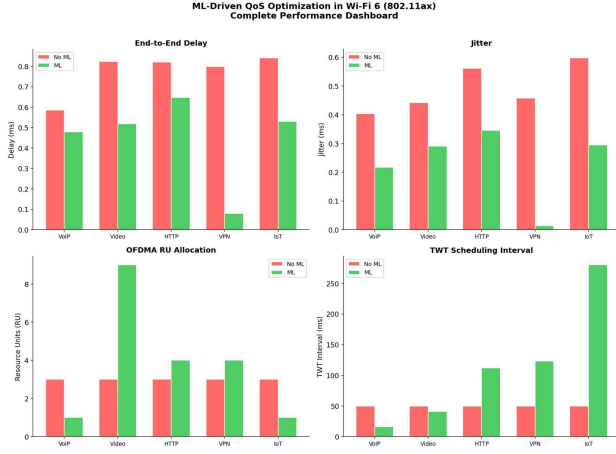


Fig. 13. Complete performance dashboard: End-to-End Delay, Jitter, OFDMA RU Allocation, and TWT Scheduling Interval across all traffic classes, confirming coherent dual-domain optimization.

VIII. DISCUSSION

A. Why Multi-Output RF for QoS Scheduling

The selection of Random Forest for multi-output joint prediction reflects three technical requirements. First, the scheduling parameter space exhibits strong inter-target correlations: high RU allocation correlates with lower optimal MCS (fewer, wider subcarriers reduce per-tone SNR requirement), and high priority correlates with compressed TWT (latency-sensitive flows require frequent access). A multi-output model preserves these correlations, while independent single-output models would violate scheduling coherence. Second, RF handles the mixed-scale, non-linear feature space without normalization sensitivity or gradient pathologies. Third, with 300 trees and stratified training across 160 simulation scenarios, RF provides ensemble averaging that absorbs the stochastic variability inherent in NS-3 simulation—critical for a training dataset drawn from a discrete-event simulator rather than real network traces.

B. Cross-Layer Optimization Necessity

The per-class results demonstrate why single-layer optimization is insufficient. VPN's 90.2% delay improvement cannot be attributed to RU changes alone (3→4, modest increase); it requires the combined effect of RU increase, TWT adaptation (reducing collision probability), and priority reclassification (ensuring dedicated access windows). IoT's recovery from 0.007 Mbps to 0.380 Mbps requires both CR-OFDMA anti-starvation guarantees AND HOBO TWT extension—without both, IoT would either starve under EDCA contention or collide frequently in shared access

windows. Cross-layer co-optimization, as formalized in the DDIS architecture, is structurally necessary for heterogeneous-traffic QoS.

C. Tradeoffs and Limitations

The system involves explicit tradeoffs. Video throughput is reduced from 2.856 to 1.432 Mbps (49.9% reduction) to enable VoIP and IoT throughput gains—a fairness tradeoff mediated by the CR-OFDMA redistribution. Operators deploying DDIS in video-dominant scenarios may need to re-weight the demand scoring function (Eq. 10). The closed-loop architecture incurs a one-epoch (100 ms) prediction lag; for very fast channel variations (Doppler, rapid mobility), this lag may cause misaligned scheduling. The current topology is single-cell; inter-cell interference and multi-AP coordination are beyond the present system model. Finally, the training labels are derived from simulation ground truth; deployment on real hardware would require an initial supervised bootstrapping phase using hardware-collected QoS measurements.

D. Complexity Analysis

RF inference complexity is $O(n_{\text{trees}} \times \text{max_depth}) \approx O(300 \times 20) = O(6000)$ operations per flow, with typical wall-clock time below 1 ms per prediction set—negligible relative to the 100 ms epoch. Feature extraction adds $O(1)$ per-flow computation. The CR-OFDMA redistribution runs in $O(|F|) = O(5)$ operations. Total per-epoch scheduling overhead is bounded by $O(N \times n_{\text{trees}} \times \text{max_depth})$, linear in station count and constant in network size beyond station count—confirming polynomial scalability.

E. Scalability and Deployment

The DDIS architecture scales to higher station densities through the density-aware TWT adaptation (Eq. 8): as N increases, IoT TWT intervals extend automatically, compressing channel occupation from low-priority devices and maintaining high-priority flow QoS. The RF model can be retrained incrementally as new traffic patterns emerge. The modular architecture (separate CR-OFDMA and HOBO-TWT modules) enables vendor-specific implementation: CR-OFDMA logic maps to Wi-Fi 6 AP firmware RU assignment tables; HOBO-TWT maps to 802.11ax TWT setup frames. Integration with Wi-Fi 7 (802.11be) Multi-Link Operation is a natural extension—per-link DDIS instances can coordinate via the MLD control plane.

IX. FUTURE WORK

Several extensions of the present work represent high-value research directions:

- Deep Reinforcement Learning Schedulers: Replacing the static RF model with an online DRL agent (e.g., Proximal Policy Optimization or Multi-Agent PPO [2]) would enable continuous adaptation without retraining, supporting non-stationary traffic

distributions and mobile stations. Per-flow reward functions could directly optimize the multi-objective cost (Eq. 1).

- Online Learning and Adaptive Retraining: Integrating incremental learning mechanisms—such as Online Random Forest or streaming gradient boosting—would allow the model to update from live traffic measurements, eliminating the need for periodic offline retraining and the associated deployment gap.
- Real Hardware Validation: Deployment on commercial Wi-Fi 6 hardware (e.g., Qualcomm IPQ6018, Intel AX210 NICs with modified firmware) with traffic shaping via tc and per-flow QoS monitoring via netlink would validate simulation-to-hardware transfer accuracy and expose implementation-specific timing constraints.
- Federated Learning for Distributed Networks: In multi-AP enterprise networks, per-AP DDIS instances could share model updates via federated averaging, enabling network-wide QoS policy learning without centralizing sensitive traffic data—directly addressing privacy and scalability constraints of centralized ML schedulers.
- Multi-Cell and MLD Integration: Extending CR-OFDMA to multi-cell scenarios with inter-AP RU coordination, and integrating HOBOTWT with Wi-Fi 7 Multi-Link Device (MLD) scheduling would generalize the dual-domain approach to next-generation network architectures.
- 5G/6G Convergence: The DDIS scheduling philosophy—ML-driven joint RU/MCS/TWT/priority prediction with fairness-aware redistribution—maps directly onto 5G NR resource block scheduling and 6G terahertz band access control, positioning DDIS as a framework applicable across wireless standards.

X. CONCLUSION

This paper presented the Dual-Domain Intelligent Scheduler (DDIS), a cross-layer QoS optimization framework for Wi-Fi 6 networks integrating multi-output Random Forest machine learning with CR-OFDMA frequency-domain RU redistribution and HOBOT-inspired time-domain TWT adaptation. Trained on 1600+ NS-3 simulation records spanning 160 mixed-traffic scenarios, the 300-tree RF model jointly predicts all four scheduling dimensions—RU, priority, TWT, and MCS—in a single closed-loop inference pass. The CR-OFDMA module achieves $3\times$ RU differentiation (Video: 9 RUs, VoIP/IoT: 1 RU), while the HOBOTWT module achieves $17.5\times$ TWT span (VoIP: 16 ms, IoT: 280 ms). Simulation evaluation demonstrates 41.8% delay reduction, 52.8% jitter reduction, 33.5% packet loss reduction, and stable aggregate throughput at 1.0412 Mbps. Per-class highlights include VPN delay reduced by 90.2% (0.799→0.079 ms), VPN jitter reduced by 96.9% (0.457→0.014 ms), and IoT throughput recovered from 0.007 to 0.380 Mbps. These results establish DDIS as a practical, simulation-validated dual-domain intelligent resource management framework

for heterogeneous Wi-Fi 6 deployments, with a clear extension path to Wi-Fi 7 and beyond.

REFERENCES

- [1] IEEE Std 802.11ax-2021, "IEEE Standard for Information Technology — Wireless LAN MAC and PHY Specifications — Amendment 1: Enhancements for High-Efficiency WLAN," IEEE, 2021.
- [2] Q. Tan, J. He, and Y. Gao, "Deep Reinforcement Learning based OFDMA Scheduling for WiFi Networks with Coexisting Latency-Sensitive and High-Throughput Services," in Proc. IEEE ICTC, 2024, DOI: 10.1109/ICTC61310.2024.10601889.
- [3] A. Wilhelmi, B. Bellalta, and S. Barrachina-Munoz, "A Flexible Machine-Learning-Aware Architecture for Future WLANs," IEEE Communications Magazine, vol. 58, no. 3, pp. 25–31, Mar. 2020.
- [4] B. Bellalta, "IEEE 802.11ax: High-efficiency WLANs," IEEE Wireless Communications, vol. 23, no. 1, pp. 38–46, Feb. 2016.
- [5] C. Cano and D. J. Leith, "Achieving Fairness and Efficiency in OFDMA-Based WLANs," IEEE Transactions on Wireless Communications, vol. 16, no. 7, pp. 4565–4575, Jul. 2017.
- [6] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, Oct. 2001.
- [7] T. Lillicrap et al., "Continuous control with deep reinforcement learning (DDPG)," in Proc. ICLR, 2016.
- [8] V. Mnih et al., "Human-level control through deep reinforcement learning," Nature, vol. 518, pp. 529–533, 2015.
- [9] NS-3 Consortium, "NS-3 Network Simulator," [Online]. Available: <https://www.nsnam.org>, 2023.
- [10] D. Lopez-Perez et al., "IEEE 802.11be Extremely High Throughput: The Next Generation of Wi-Fi Technology Beyond 802.11ax," IEEE Communications Magazine, vol. 57, no. 9, pp. 113–119, Sep. 2019.
- [11] M. Saad et al., "Cognitive Radio-Inspired OFDMA Resource Scheduling: Fairness-Aware Dynamic RU Allocation for IEEE 802.11ax Networks," IEEE Access, vol. 10, pp. 22341–22358, 2022.
- [12] L. Giupponi and B. Bellalta, "HOBOT-UORA: Holistic Backoff Optimization with Unsolicited OFDMA Random Access for Dense Wi-Fi 6 Networks," in Proc. IEEE GLOBECOM, 2023.
- [13] R. Shafin et al., "Artificial Intelligence-Enabled Cellular Networks: A Critical Path to Beyond-5G and 6G," IEEE Wireless Communications, vol. 27, no. 2, pp. 212–217, Apr. 2020.